TEXORA · LECTURE SHEET

Design of 3D Machine Parts

Parametric CAD modelling in practice (SOLIDWORKS and Blender workflows): sketches and constraints, the core features — extrude, revolve, sweep, loft — patterns, assemblies and mates, and the design intent that makes a model editable instead of fragile.

ME-1202 · ENGINEERING DRAWING-II · TOPIC 1 · LEVEL 1 TERM 2

1. Parametric modelling — the model is a recipe

In **parametric CAD** a part is not a frozen lump; it is an ordered **feature tree** — a recipe of sketches and operations driven by dimensions. Change the sketch dimension “shaft Ø40” to Ø45 and every dependent feature (keyway, bearing seat, the assembly it sits in, the drawing) updates. This only works if the model is built with **design intent**: relationships captured deliberately — symmetry about a centre plane, holes concentric with a boss, a pattern driven by “6 × equally spaced” rather than six separately placed holes that must each be edited by hand. A model that rebuilds cleanly after a dimension change is a professional model; one that collapses was merely drawn.

2. Sketches — where parts are really designed

Every solid begins as a 2D **sketch** on a plane or face. Sketch geometry is tamed by **constraints** (coincident, concentric, parallel, perpendicular, tangent, symmetric, equal) and **dimensions**. A sketch is **fully defined** when constraints + dimensions leave zero degrees of freedom — the professional standard, because an under-defined sketch shifts silently when edited. Good practice: anchor the sketch to the origin or centre planes, exploit symmetry (sketch half, mirror), keep sketches simple and put detail (fillets, chamfers) in later features, and name important dimensions.

3. The core solid features

FEATURE	WHAT IT DOES	TYPICAL MACHINE PARTS
Extrude (boss/cut)	Pushes a profile linearly; blind, through-all, up-to-face, mid-plane	Plates, brackets, flanges, keyways (cut)
Revolve	Spins a profile about an axis — one sketch makes any turned part	Shafts, pulleys, bushes, bobbins and cones for winding

Sweep	Drags a profile along a path	Pipes, springs (helical path), handles, yarn guides' curved tubes
Loft	Blends between two or more profiles of different shape	Transition ducts, impeller passages, ergonomic housings
Fillet / chamfer	Rounds or bevels edges — applied late, largest first	Stress relief at shaft shoulders; lead-ins on bores
Patterns & mirror	Linear/circular repetition of a feature	Bolt-hole circles (e.g. 6 × Ø9 on a 100 mm pitch circle), gear teeth, vents
Shell / rib / draft	Hollows a solid to a wall thickness; adds stiffening webs; tapers faces for mould release	Cast housings, covers, moulded parts

WORKED EXAMPLE — MASS PROPERTIES THE CAD COMPUTES FOR FREE

Model a plain steel shaft: Ø40 mm × 200 mm, revolved in one feature. Volume $V = (\pi/4) \times 40^2 \times 200 = 251{,}327\ \text{mm}^3 \approx 251.3\ \text{cm}^3$. With steel $\rho = 7.85\ \text{g cm}^{-3}$, mass $= 251.3 \times 7.85 \approx$ **1.97 kg**. Its mass moment of inertia about the shaft axis, $I = \tfrac{1}{2}mr^2 = \tfrac{1}{2} \times 1.973 \times (0.020)^2 \approx$ **3.9×10^{-4} kg·m²** — exactly the figures the CAD mass-properties panel reports, and the starting point for any drive or balancing calculation (compare PHY-1103).

4. Assemblies and mates

Parts meet in an **assembly**, positioned by **mates** (SOLIDWORKS) or constraints (Blender rigs/collections for the textile models of Topics 3–4): **coincident** (faces together), **concentric** (shaft in bore), **distance/angle**, **tangent**, **parallel/perpendicular**. Each mate removes degrees of freedom; a well-mated component keeps only the motion it should have — a pulley keeps one rotation, a sliding cover one translation. Standard hardware comes from libraries (Toolbox: bolts, nuts, bearings, keys) rather than being remodelled, and interference detection plus motion studies in the assembly catch design errors while they are still free to fix.

5. From model to manufacture

The 3D model is the master: 2D drawings are generated views of it (Topic 2 of ME-1102's companion skills — dimension once, from the model, and the drawing updates with it); tolerances and fits are assigned by function (a bearing seat is not dimensioned like a cover plate); and the model exports to manufacturing — STL for 3D printing, STEP/IGES for exchange, DXF for cutting. Design rules from the workshop floor still apply inside the computer: avoid knife edges, give internal corners the radius a

cutter can make, keep wall thickness uniform in cast and moulded parts, and model the part in the orientation it will be machined or printed.

6. Key points

- Parametric = feature tree + driving dimensions; design intent is what survives an edit.
- Fully defined sketches only: constraints first, dimensions second, anchored to origin/centre planes.
- Extrude, revolve, sweep, loft build almost everything; fillets and patterns come late; standard parts come from libraries.
- Mates remove degrees of freedom deliberately — leave exactly the motion the mechanism needs.
- The model is the single source for drawings, mass properties, simulation (Topic 2) and manufacturing files.

7. Practice

Answer in your own words; brief model answers follow.

1. A flange needs 6 bolt holes of Ø9 mm equally spaced on a 100 mm pitch circle. Describe the efficient modelling strategy and the design-intent failure of modelling six independent holes.
2. A stepped shaft (Ø50 × 60 mm, Ø40 × 80 mm, Ø30 × 40 mm) is to be modelled. Which single feature builds it, and what does the sketch contain?
3. Calculate the mass of a steel bush (outside Ø60 mm, bore Ø40 mm, length 50 mm; $\rho = 7.85$ g cm^{-3}) — the check you would run against the CAD mass readout.
4. An assembly's pulley, mated concentric to its shaft and coincident to a shoulder, still spins and cannot slide. How many degrees of freedom remain, and is that correct for a pulley keyed to the shaft? What changes for a free-spinning idler pulley?
5. A colleague's model collapses when the boss diameter is enlarged: the hole that was "in the middle" stays behind and the fillet fails. Diagnose the two design-intent faults.
6. Why are fillets normally added near the end of the feature tree rather than inside the base sketch?

Answer key

1. Model one hole (an extruded cut) positioned by the pitch-circle radius, then a circular pattern: count 6, spacing "equal, 360°". Editing the count or the pitch diameter regenerates all six. Six independent holes carry six copies of every dimension — change the pitch circle and five stay behind; a hole can be mis-typed without any warning. The pattern encodes the intent "six equal holes on this circle" once.

2. Revolve. The sketch is the half-profile (a stepped rectangle outline) drawn on a plane through the axis, with the axis as a construction/centre line; one revolve of 360° creates all three diameters in a single feature — mirroring how the part is actually turned on a lathe.
3. Annulus area = $(\pi/4)(60^2 - 40^2) = (\pi/4)(2{,}000) = 1{,}571$ mm². Volume = $1{,}571 \times 50 = 78{,}540$ mm³ = 78.5 cm³. Mass = $78.5 \times 7.85 \approx 617$ g ≈ 0.62 kg.
4. A free rigid body has 6 DOF (3 translations, 3 rotations). Concentric removes 2 translations + 2 rotations; coincident on the shoulder removes the axial translation: 1 DOF remains — rotation about the shaft axis. For a keyed pulley even that must go (in reality the key fixes it; in CAD, lock rotation or add the key/keyway mate) — a keyed pulley should end fully defined. A free idler is correctly left with exactly that 1 rotational DOF: the mate scheme deliberately preserves its working motion.
5. (1) The hole was dimensioned from an edge, not constrained concentric to the boss — placement was numerical coincidence, not intent; when the boss grew, the centre moved but the hole's fixed edge-distance did not. Fix: a concentric constraint to the boss's circular edge/axis. (2) The fillet was sized/applied so that enlarging the boss left no face for it, or it was placed early and later features depended on it — fillets belong late in the tree, applied to final edges, so upstream size changes cannot orphan them.
6. Sketch fillets freeze radii into the driving profile, complicate the sketch, and break when faces they reference change size or vanish; feature fillets sit at the end of the recipe where they simply re-apply to whatever edges exist after all size edits. They can also be suppressed for simulation or machining models (Topic 2) without touching the base geometry — small radii barely affect stress results but heavily tax meshing.
